

Experimental Setup

Does the SS OCF 1 action implementation preserve its chosen common principal structure on smooth local backgrounds?

Experiment: gross.continuum-action.v1 Model: signal-space.ss-ocf-1.action-audit.v1
Runtime configuration: fixtures/research/gross-test-05.json
Plan lock: 7dcbb82ce275c4c30fa92c5a4f6335a40fc83b2b3bcdecdf927d51bf116b3915

MODEL AND NUMERICAL SETUP

Action: SS OCF 1 Einstein plus four real q fields, neutral a with $Z=1+0.1s$ and real χ ; potentials per program section 7.3. Separate quartic orientation action control.
Units: $c=\hbar=m=M_0=1$, $(+---)$, $|k|=1$; fields mass dimension 1, first derivatives mass^2 .
Domain: 200 seeded invertible smooth local coframes, arbitrary local q,a,χ gradients and symmetric second jets; 20 deterministic sphere directions.
Boundaries: Frozen pointwise principal-symbol/action audit; no spatial boundary, PDE mesh or evolution. Gravity flat linearized harmonic TT symbol only.
Initial Data: PCG64 seed 2026092505, lapse $.8..1.2$, shift $\pm .2$, spatial coframe $I \pm .15$; fields ± 1 , gradients/jets $\pm .5$.
Solver: Action polarization on all 24 derivative slots, independent complex-step field forces, saved roots and independent null-polynomial/coframe audit; TT Einstein symbol and constraints.
Tolerances: All normalized errors $< 1e-10$; positive kinetic eigenvalues; control split $> 1e-3$; resource 150s/512MiB/24MiB.

COMPETING PREDICTIONS

common: $H=\text{diag}(1,1,1,1,Z,1)$ tensor g^{-1} ; all six roots agree, valid $Z>0$.
split: Quartic orientation control splits a physical speed; negative Z violates positive kinetic health.

CONTROLS

negative- z : detect failed kinetic health; method: set $Z=-0.5$ in an explicitly artificial frozen control
quartic: detect a material cone split; method: separate versioned $-\kappa H_{ab} H^{ab}/4$ orientation action with $p=.6, \kappa=.4$
complex-step: independently verify written equations; method: complex-step differentiation of complete matter L with nonzero q,a,χ jets

PREREGISTERED DECISIONS

coverage (full coverage)
Pass: exactly $200 \times 20 \times 6 \times 2$ finite roots and complete arrays
Fail: complete evidence contradicts the preregistered condition
Unresolved: missing, nonfinite or incomplete evidence prevents assessment
Uncertainty: floating point and coframe conditioning; no stochastic confidence or global PDE convergence claim

Experimental Setup

hessian (normalized action Hessian residual)

Pass: maximum scale-normalized action tensor and operator-metric error $<1e-10$

Fail: complete evidence contradicts the preregistered condition

Unresolved: missing, nonfinite or incomplete evidence prevents assessment

Uncertainty: floating point and coframe conditioning; no stochastic confidence or global PDE convergence claim

equations (Euler-Lagrange local-jet residual)

Pass: maximum scale-normalized action/equation discrepancy $<1e-10$

Fail: complete evidence contradicts the preregistered condition

Unresolved: missing, nonfinite or incomplete evidence prevents assessment

Uncertainty: floating point and coframe conditioning; no stochastic confidence or global PDE convergence claim

characteristics (six-sector null-root residual)

Pass: maximum root equality and null polynomial residual $<1e-10$

Fail: complete evidence contradicts the preregistered condition

Unresolved: missing, nonfinite or incomplete evidence prevents assessment

Uncertainty: floating point and coframe conditioning; no stochastic confidence or global PDE convergence claim

positive-kinetic (valid and negative-control kinetic eigenvalues)

Pass: all valid field eigenvalues >0 and negative $Z=-0.5$ is detected

Fail: complete evidence contradicts the preregistered condition

Unresolved: missing, nonfinite or incomplete evidence prevents assessment

Uncertainty: floating point and coframe conditioning; no stochastic confidence or global PDE convergence claim

orientation-control (quartic-control second speed)

Pass: separate quartic action speed <0.999 with independent formula error $<1e-10$ and positive coefficients

Fail: complete evidence contradicts the preregistered condition

Unresolved: missing, nonfinite or incomplete evidence prevents assessment

Uncertainty: floating point and coframe conditioning; no stochastic confidence or global PDE convergence claim

gravity-symbol (harmonic-gauge TT symbol and linearized constraint residual)

Pass: linearized harmonic TT symbol and flat linearized Hamiltonian/momentum constraints $<1e-10$

Fail: complete evidence contradicts the preregistered condition

Experimental Setup

Unresolved: missing, nonfinite or incomplete evidence prevents assessment

Uncertainty: floating point and coframe conditioning; no stochastic confidence or global PDE convergence claim

VISUAL QUESTIONS

characteristic-surfaces [evidentiary]: Do all six matter sectors share one coframe null cone in shifted coordinates?

View: static figure and exact plot data; observables: Do all six matter sectors share one coframe null cone in shifted coordinates?

common: all valid matter branches share metric null roots and positive kinetic matrix

split: orientation quartic has separate characteristic; negative Z fails health

Controls: negative-z; uncertainty: full arrays retained; flat local jets and fixed directional samples limit scope

Data: reports/<report-id>/plot-data/characteristics.csv; transforms: display four coframes in root plot; all 200 in residual plot; no fit or smoothing; log display floor 1e-18

action-audit [evidentiary]: Do action derivatives agree with the written equations on nonzero jets?

View: static figure and exact plot data; observables: Do action derivatives agree with the written equations on nonzero jets?

common: all valid matter branches share metric null roots and positive kinetic matrix

split: orientation quartic has separate characteristic; negative Z fails health

Controls: complex-step; uncertainty: full arrays retained; flat local jets and fixed directional samples limit scope

Data: reports/<report-id>/plot-data/samples.csv; transforms: display four coframes in root plot; all 200 in residual plot; no fit or smoothing; log display floor 1e-18

failure-controls [evidentiary]: Do unhealthy Z and quartic orientation change distinct accepted properties?

View: static figure and exact plot data; observables: Do unhealthy Z and quartic orientation change distinct accepted properties?

common: all valid matter branches share metric null roots and positive kinetic matrix

split: orientation quartic has separate characteristic; negative Z fails health

Controls: negative-z, quartic; uncertainty: full arrays retained; flat local jets and fixed directional samples limit scope

Data: reports/<report-id>/plot-data/controls.json, reports/<report-id>/plot-data/summary.json; transforms: display four coframes in root plot; all 200 in residual plot; no fit or smoothing; log display floor 1e-18

RESOURCE LIMITS

{"max_memory_mb": 512, "max_output_mb": 24, "max_wall_seconds": 150}

Experimental Setup

PROHIBITED INFERENCES

The common cone is chosen by the action and is no proof of emergent spacetime or an operational detector geometry.

Flat TT constraint identities do not establish nonlinear Einstein evolution or constraint-satisfying initial data.

The orientation failure control is a different model and does not make SS OCF 1 topologically protected.